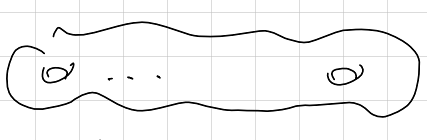
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



$(n\geq 1$

$\downarrow$

$\pi_1=0$.

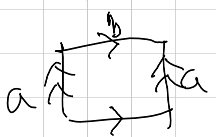
2-dim mfd

$L(p,q)$

$M\# N$. $\wedge$ $\vee$

$D^1\times S^1/e^{2\pi i p}, e^{2\pi i q}$

$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.

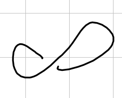
klein. free product.

~~$\mathbb{Z}\times\mathbb{Z}$~~

ababab b..

$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e$.

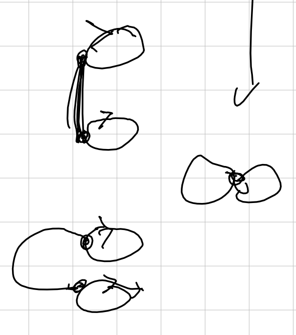


$\langle a,b \mid ab=ba\rangle$

$\mathbb{Z}\times\mathbb{Z}$

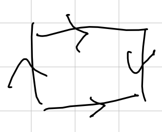
$S^1\vee S^1$

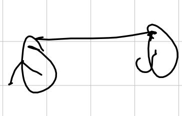
$ab=ba \to$ group $\mathbb{Z}$.



$S^1\vee S^2$

$T: \mathbb{Z}\times\mathbb{Z}$





$\frac{S^1\vee S^1}{\mathbb{Z}*\mathbb{Z}}$

$T^1: \mathbb{Z}*\mathbb{Z}$

$K^1: \mathbb{Z}*\mathbb{Z}$

theorem

Theorem

$S^1\vee S^1$

$\mathbb{Z}\times\mathbb{Z}$

S^1

$\updownarrow$

$\mathbb{Z}$

$S^3:0$.

$\downarrow$

$\mathbb{R}^3:0$.

M: n-dim

with $(n\geq 3)$

then $\pi_1(M)\simeq\pi_1(M\setminus\bullet)$.

$\simeq\pi_1(M\setminus 2pt)$

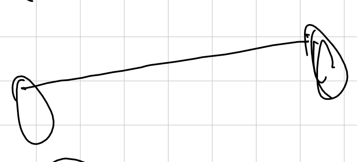
$\simeq\pi_1(M\setminus\{x_i\})$.

$\pi_1(S^n)=0$

$n\geq 2$.

$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$

$(n\geq 2)$.

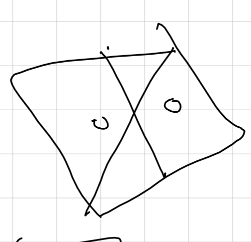


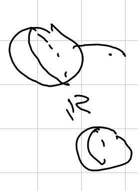
~~tree~~

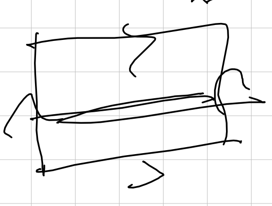
$S^1\vee S^1\vee\mathbb{R}$.

$\downarrow$

$\mathbb{Z}*\mathbb{Z}$

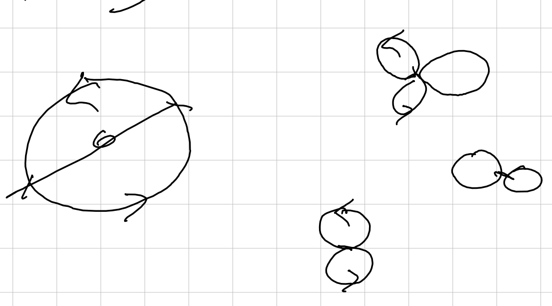




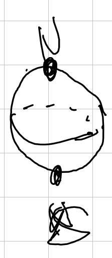


$: S^1\vee S^1\vee S^1$

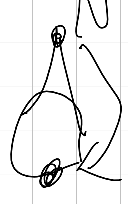
$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

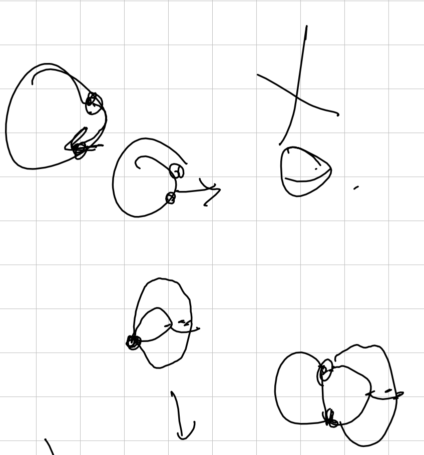


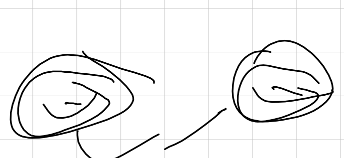
Exemple.



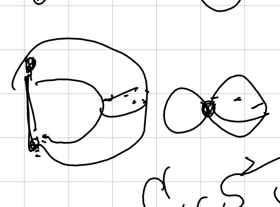
$S^2/N=S$ N



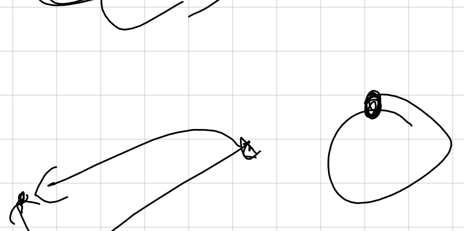




$\to\pi_1(\ldots)$



$S^1\vee S^2$



S^1: 0-cell 1개

1-cell 1개

S^2: 0-cell 1개

2-cell 1개

$S^1\vee S^1$

$\mathbb{RP}^2$: 0-cell: 1개

1-cell: 1개

2-cell 1개

$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba\rangle$

$(\mathbb{Z}*\mathbb{Z})/\alpha$

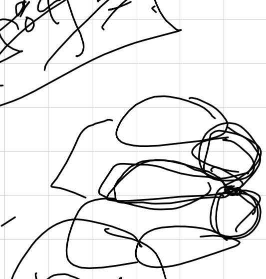
$\mathbb{RP}^2: \mathbb{Z}_2$ $\langle a \mid a^2\rangle$

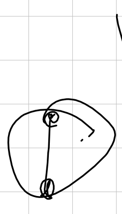
$a^2=1$

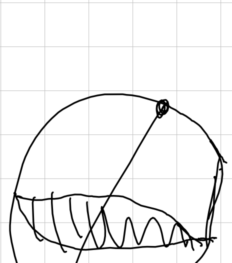
~~$\mathbb{RP}^2$: 0-cell 1개 1-cell 1개 2-cell 1개~~

$\pi_1(\mathbb{RP}^2)=\pi_1(\mathbb{RP}^2)$

$\cong\mathbb{Z}_2$.





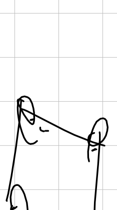


$\leftarrow$ $x^2+y^2+z^2=1$

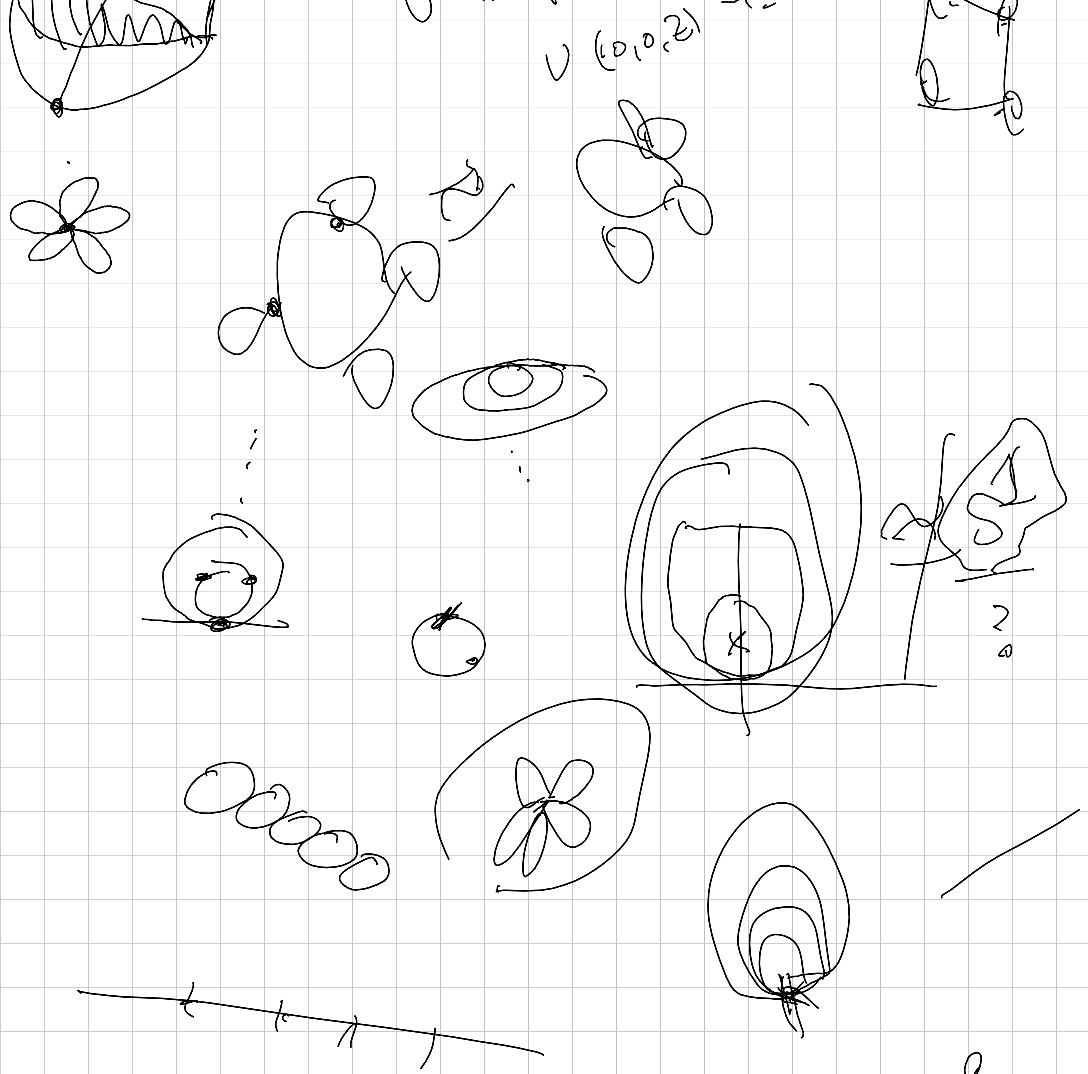
$\cup$ $x^2+y^2\leq 1$

$\cup$ $(0,0,z)$ $-1\leq z\leq 1$.

$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$



$\mathbb{R}/\mathbb{Z}$

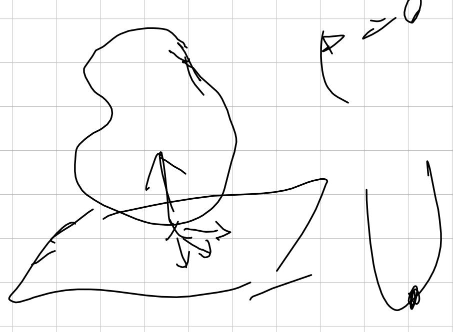


$Z=x^4+y^4$

$\downarrow$ exp

$K_1=f(\ldots)$

$K_2=f(\ldots)$



$r=0$

$\frac{1}{r^n+z}$.

$\int\frac{Rdz}{y}\int\frac{1}{z}dz$

$2\pi i$

$2\pi i\times(\ldots)$

0